

What are we investigating?

ENSO, clouds and your mini research project

YOU LEAVE WITH

The notebook running, your first cloud figures, and two patterns worth a closer look.



Tutorial 1 of 8 • Semester 1

Today's plan

Goal: Get the notebook running and see your first cloud figures.

- 1 The project: one question, 3–5 figures, 2,000 words
- 2 The two ingredients: ERA5 reanalysis and ENSO
- 3 Why should clouds change with ENSO?
- 4 A tour of the notebook
- 5 Get it running and note two patterns

BRING / HAVE READY

- A laptop (or a cluster PC)
- The Binder link from Canvas
- The student handbook
- Somewhere to save files: Binder sessions are temporary



The bigger picture: Earth's clouds from space

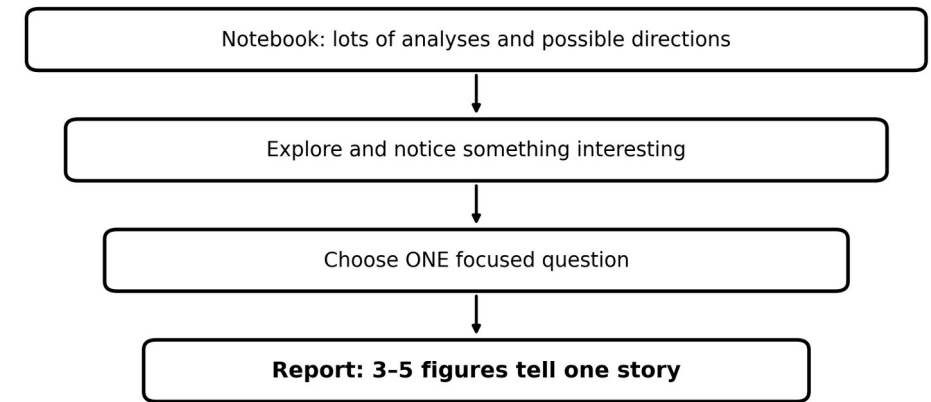
- Cloud bands like these **shift with ENSO**, and with them the odds of drought and flood across much of the planet.
- ENSO is the largest source of **year-to-year** climate variability on Earth.
- The signals you'll analyse are the same ones national weather services watch for **seasonal forecasts**.
- This term you'll find one of those signals in real data and explain it: a small, genuine piece of climate science.



A mini research project, not a checklist

- The notebook contains **more analyses than you can use**. That's deliberate: it's a toolbox.
- **Explore** first. Then choose one aspect of ENSO and clouds that interests you.
- Answer **one focused question** with about **3-5 figures**, in a report of at most **2,000 words**.
- You are **not** expected to write the Python from scratch, or to understand every line.
- Report: **100%** of the course mark. AI category: **AI Minimal** (more next week).

The report is not a record of everything you did



The year at a glance

DATE	TUTORIAL	YOU PRODUCE
Thu 1 Oct	Launch: ERA5, ENSO, notebook running	Two patterns
Thu 8 Oct	Readers and the report • ERA5 cloud, weighting, anomalies • finding sources	150 words; reference manager
Thu 15 Oct	The Introduction • thresholds and composites	Purpose statement
Thu 29 Oct	Commit to ONE question • correlation, choosing a region	Storyboard; Methods draft
Thu 12 Nov	Figures as evidence • significance: r , n , p	Results draft
Thu 19 Nov	Paragraphs that argue • reporting statistics	Introduction draft
Thu 26 Nov	Discussion and Conclusion • lags and mechanisms	Discussion draft
Thu 3 Dec	Draft complete: from writer to editor	Complete first draft

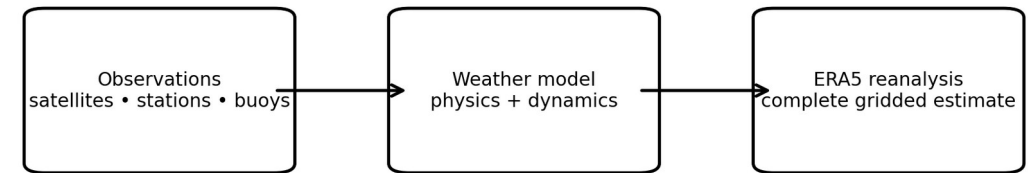
Then: draft submitted **Fri 5 Feb** (formative; no DASS extensions) → peer feedback (w/c 8 or 15 Feb) → group presentation (w/c 1 Mar) → final report **Fri 12 Mar, 12:00**.

METHODS

Ingredient 1: ERA5 reanalysis

- Observations are **incomplete** in space and time.
- A **weather model** represents the physics and dynamics of the atmosphere.
- **Data assimilation** blends the two into a complete, gridded, physically consistent record: a **reanalysis**.
- **Ours:** ERA5 monthly means, 0.75° grid, **January 2005 – August 2025**, including total, low, medium and high cloud cover.
- Reanalysis is neither a plain archive of observations nor a free-running climate model. (Next week: what that means for **cloud**.)

Data assimilation combines observations with a numerical model



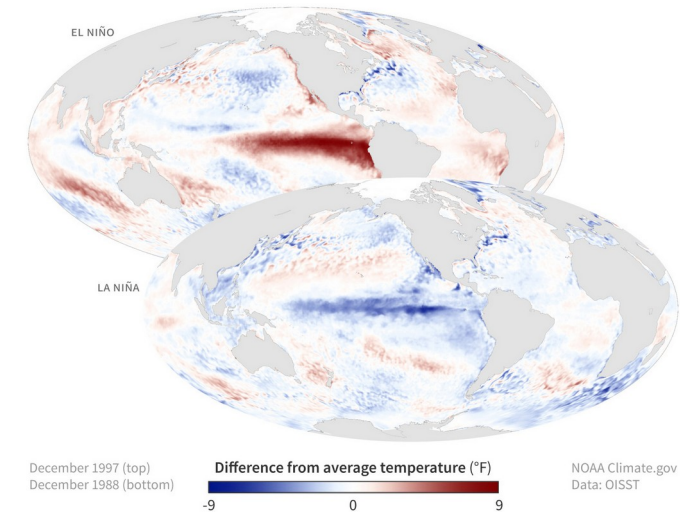
Reanalysis is neither a simple observation archive nor a free-running climate model.

METHODS

Ingredient 2: ENSO

- **El Niño:** an anomalously **warm** central and eastern tropical Pacific. La Niña: anomalously **cool**.
- The ocean pattern drags **winds, convection and rainfall** with it, so cloud is a natural response to study.
- We measure ENSO with the **ONI**: the 3-month running-mean SST anomaly in the Niño 3.4 region.
- **In our period:** El Niño in 2009–10, **2015–16** (very strong) and 2023–24; La Niña in 2010–12 and 2020–23.

ONI TEMPERATURE PATTERNS



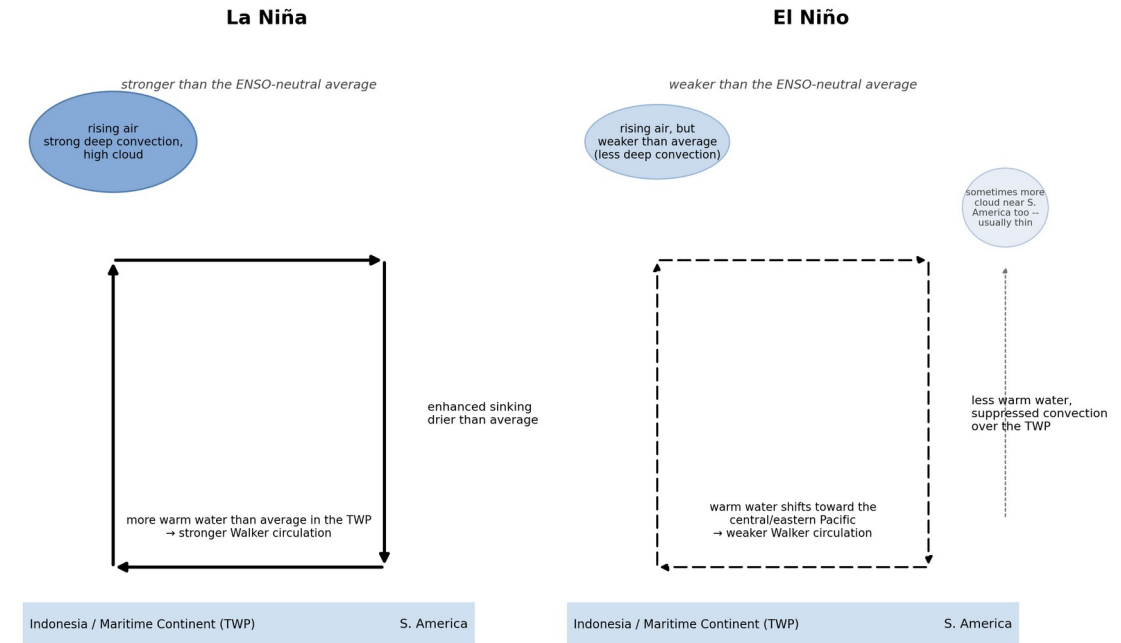
METHODS

Why should the clouds change?

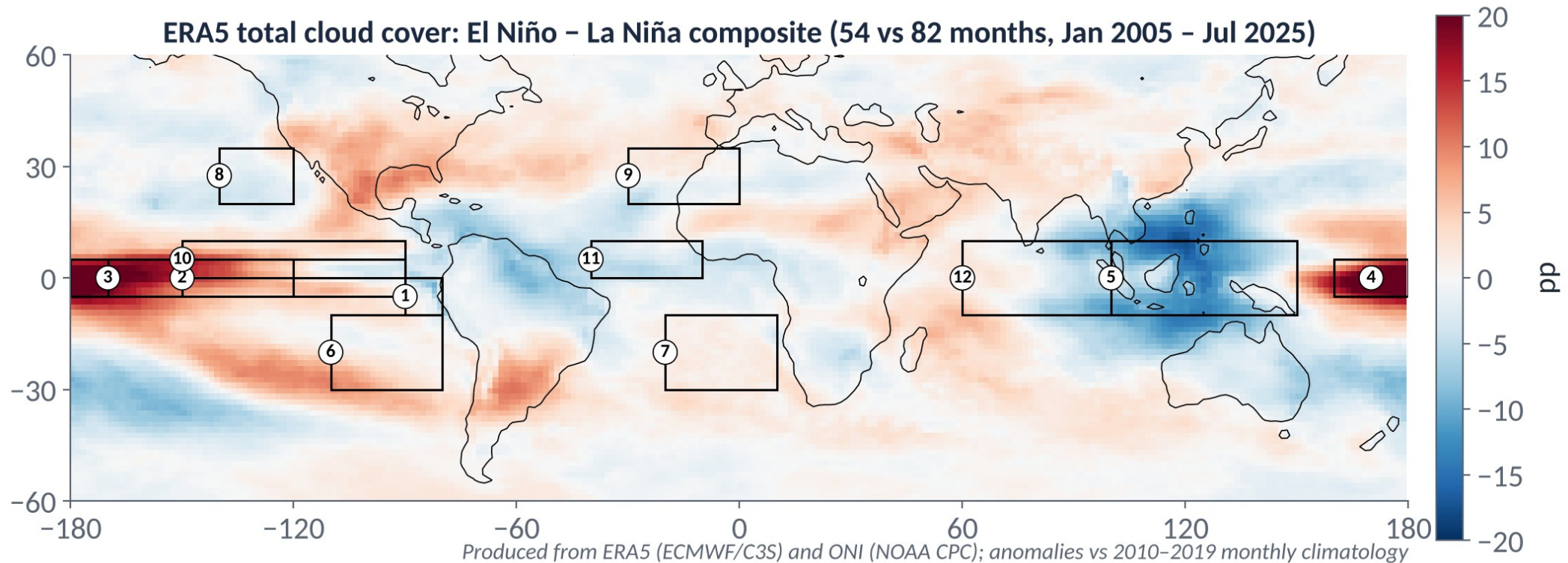
- Warm sea surface and the atmospheric circulation are **coupled**.
- The Maritime Continent stays the Walker cell's **rising branch**: El Niño **weakens** the circulation and La Niña **strengthens** it.
- During El Niño, convection and high cloud **shift towards the central Pacific**, and are suppressed over the Maritime Continent.
- Any extra cloud further east near South America is usually **thin**, not a new deep-convection centre.
- **Low cloud** can respond through different physics (stability, subsidence). Keep an open mind.

Why ENSO changes the strength of tropical cloud

The Maritime Continent is always the rising branch and the eastern Pacific the sinking branch. El Niño weakens that circulation (and can add thin cloud further east); La Niña strengthens it.



By December you'll be able to explain this map



- Total cloud in El Niño months minus La Niña months, from **the same ERA5 data you'll use**. Red = cloudier during El Niño; blue = cloudier during La Niña.
- The numbered boxes are the notebook's regions. Your report will explain **one** part of a map like this, in depth.

A tour of the notebook

0 • 0b	Setup and load the data (run once)	CORE
1	The global seasonal cycle	CORE
2	Global ENSO maps: composite and correlation	CORE
3 (+3b)	Global-mean anomalies (+ year-to-year tests)	CORE (3b optional)
4	Regional analysis: 12 numbered regions	EXPLORE
5	ONI vs regional anomaly: correlation and regression	EXPLORE
6	Autocorrelation and lag	OPTIONAL
STOP	Choose your investigation (before more analysis)	

0b. Load the data (run once)

CORE — This is the only cell that reads the NetCDF files, and it may take a minute or two the first time. Every later section reuses what is loaded here:

- `df_globals` — global area-weighted monthly means of TCC, TCLW, TCIW, TCWV
- `tcc_grid, lat, lon, time_index` — the full (time, lat, lon) TCC field, used for maps
- `oni` — the ONI index, aligned to a (year, month) index

If this cell raises `FileNotFoundError`, check that the ERA5 files are in `./era5_monthly_nc/` (see the project README).

Found 21 monthly ERA5 files (2005-2025). Reading...

Loaded global means for 231 months, grid shape (252, 37, 72) (time, lat, lon), ONI for 252 months.

What '0b. Load the data' looks like when it works

THREE RULES

- Run **from the top, in order**: later cells reuse data loaded in 0b.
- Binder sessions are **temporary**: download figures/, results/ and your notebook.
- To study another variable, change **one line**: `VARIABLE = "lcc", "hcc", ...`

How your report will be marked

40–49%

Pass

Question present but vague; figures not well chosen; reads like a notebook summary rather than an argument.

50–59%

2:2

One reasonably specific question, answered with thin reasoning; figures mostly relevant; attribution present.

60–69%

2:1

A focused question, directly answered; 3–5 well-designed figures; quantified, physically interpreted evidence.

70–79%

First

Well-chosen question; real physical insight; Claim → Evidence → Interpretation → Qualification throughout; independent thinking in the Discussion.

80%+

Exceptional

All of the above plus something beyond what was taught: an original angle or a careful treatment of a subtlety. Rare.

Full descriptors are in the handbook (§9). The same scheme applies all year, so you know from day one what 'good' looks like.

IN-CLASS ACTIVITY • 18 MIN

Get something working

- 1 Open the project through the **Binder link** on Canvas (the first launch can take a few minutes).
- 2 Run the notebook **from the top**, through **0b. Load the data**, then section **1**, and section **2** if time allows.
- 3 Look at the first cloud figures. Where is cloud cover high or low? What changes through the year?
- 4 Write down **two patterns or regions** that catch your eye, with **where, what and when**.
- 5 Before you leave: **download** your figures and notebook. Binder forgets everything.

YOU LEAVE WITH

- The notebook running (or a clear note of what went wrong)
- Two patterns written down

PAIR CHECK

- Compare your two patterns with a neighbour. Did you notice the same things?
- If your Binder fails, pair with someone whose works and run it together.

Before Tutorial 2 · Thu 8 Oct

About 1½ hours. Next week the writing starts, using what you notice now.

1A · 45 min

Finish running the notebook

- Sections **0–3** at least
- Save figures **outside Binder**
- Note anything that errors

1B · 15 min

Two patterns, written properly

- For each: **where**, **what** (which cloud variable), **when** (season / ENSO phase)
- One sentence on **why** it might happen

1C · 30 min

Read the handbook

- **§1–3**: the project, ERA5, ENSO and clouds
- Skim §9 (marking)
- Download the **writing workbook** from Canvas

NOTEBOOK STRAND

- Sections 0–3. Analysis slides: **01 Project launch** (and 02 if you're keen).

READ · SCHULTZ (2009) ELOQUENT SCIENCE

- Optional: the **Introduction** ('An incoherent truth') and '**How to use this book**': a quick taste of the course writing text

Sources and attribution

- **ERA5:** ECMWF / Copernicus Climate Change Service. Hersbach, H. et al. (2020) The ERA5 global reanalysis. Q. J. R. Meteorol. Soc., 146, 1999–2049.
- **ONI:** NOAA Climate Prediction Center.
- **ENSO maps:** NOAA Climate.gov, based on NOAA OISST data.
- **Blue Marble:** NASA/NOAA/GSFC/Suomi NPP/VIIRS, Norman Kuring (4 Jan 2012).
- **Composite map:** produced for this course from ERA5 and ONI (anomalies relative to the 2010–2019 monthly climatology).
- **Schematics:** created by the course team for these tutorials.